Bitcoin Price Prediction

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Problem Statement

Bitcoin Price Prediction

Why?

Bitcoin prices are highly volatile, influenced by multiple market factors, and difficult to interpret using raw historical data alone. Most analysts and retail investors rely on fragmented tools for charting, technical indicators, and forecasting, which makes analysis inconsistent, time-consuming, and hard to reproduce. This creates a need for a unified system that can both explain past market behavior and generate reliable short-term predictions.

What?

The problem is to design and build an end-to-end analytics platform that integrates historical Bitcoin data ingestion, interactive visualisation, technical indicator analysis, feature engineering, and predictive modeling in a single workflow. The system should help users explore trends, evaluate signals, and make data-driven decisions while handling noisy time-series data, minimising prediction error, and presenting results in an interpretable way.

Why This Matters?

This matters because decisions in crypto markets are often made under uncertainty, and small mistakes can lead to significant financial losses. A unified platform for Bitcoin analysis and prediction helps transform scattered, hard-to-interpret data into clear insights that users can act on with greater confidence. By combining visualisation, technical indicators, and forecasting in one place, it reduces analysis time, improves consistency, and makes advanced methods accessible beyond expert traders. It also supports more transparent, reproducible decision-making, which is essential for building trust in data-driven investing workflows.

Dataset Overview

Bitcoin Dataset Features

Item	Detail
Training rows	1556
Test rows	7
Columns	Date, Open, High, Low, Close, Volume, Market Cap
Split	Chronological 80/20

Date	Open	High	Low	Close	Volume	Market Cap
Jul 31, 2017	2763.24	2889.62	2720.61	2875.34	860,575,000	45,535,800,000
Jul 30, 2017	2724.39	2758.53	2644.85	2757.18	705,943,000	44,890,700,000
Jul 29, 2017	2807.02	2808.76	2692.8	2726.45	803,746,000	46,246,700,000
Jul 28, 2017	2679.73	2897.45	2679.73	2809.01	1,380,100,000	44,144,400,000

Our Approach: Ridge Regression

Ridge regression with L2 regularization came out to be the most robust and consistent performer out of various other machine learning models tested for this dataset, which will be outlined in our experiments in the following slides.

The final model uses 4 input features : 1-day percentage return of Close, 3-day percentage return of Close, relative deviation from 7-day moving average and 7-day rolling standard deviation of 1-day percentage return of Close.

Metrics?

RMSE

Measures prediction error in actual USD terms. It penalizes large "misses" more heavily, providing a clear view of the model's reliability and potential financial risk during volatile periods.

Directional Accuracy

Focuses on market trends (Up/Down) rather than exact price. It validates the model's utility as a signal generator, proving it can effectively capture profitable movements for active trading.

Annualised Sharpe Ratio

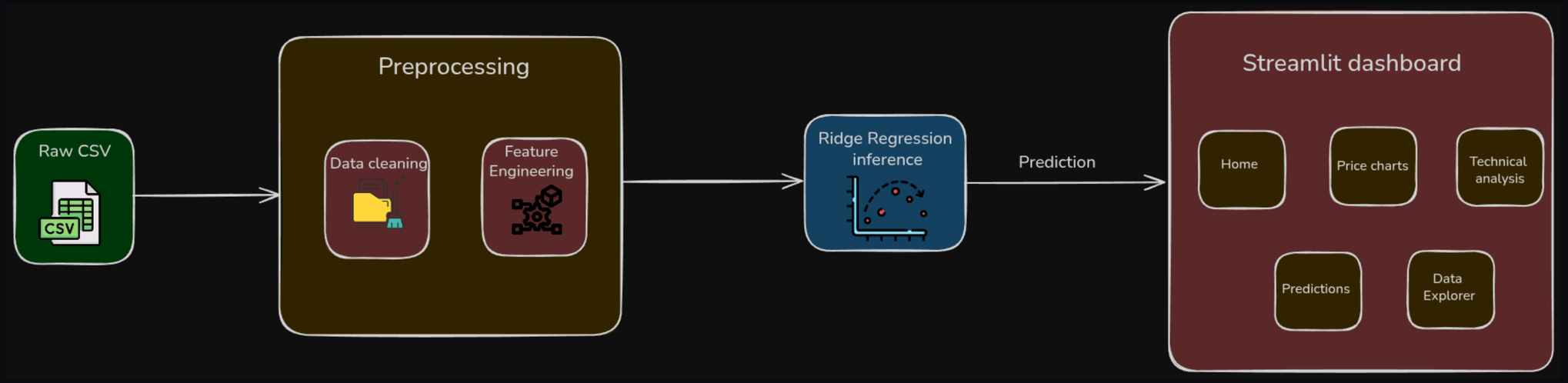
Evaluates risk-adjusted performance by comparing hypothetical gains to market volatility. It is the gold standard for determining if a model's edge is consistent or simply a result of high-risk exposure.

Why This Model?

- Target is noisy by nature : variance control is more important than raw model flexibility
- High regularisation matches the design goal: robustness over aggressiveness
- Preserves all domain-selected signals instead of dropping them
- Solves the problem of unstability due to multicollinearity : The L2 term shrinks coefficients smoothly, reducing sensitivity to sampling noise and collinear inputs

Ridge was chosen because it offers the best bias-variance tradeoff for a noisy, collinear, time-ordered return-forecasting problem, while remaining interpretable, stable, and production-friendly.

Methodology: Final Pipeline



This end-to-end architecture streamlines the transition from raw CSV ingestion to live inference. The pipeline implements a systematic preprocessing stage for data cleaning and feature generation, followed by a Ridge Regression engine to forecast daily returns. The results are integrated into a multi-page Streamlit application, providing robust visualisation of price trends, model performance, and data exploration tools.

Methodology: Preprocessing

Cleaning and Standardisation

Converting currency strings (Volume/Market Cap) into numeric formats. Dates were parsed and sorted chronologically to ensure the integrity of the time-series structure, which is vital for sequential prediction.

Mometum and Trend Indicators

Engineered lagged return features (Return_1d, Return_3d) to capture short-term price momentum and historical velocity. A 7-day Simple Moving Average difference (SMA_7_diff) was added to identify price deviations from the trend, providing a signal for potential mean reversion.

Volatility and Target Alignment

Calculated a rolling 7-day standard deviation (Volatility_7) to quantify market risk and noise levels at any given time. The target variable (Next_Return) was then aligned using a backward shift, ensuring the model learns to predict future returns without any data leakage.

Before

	Date	Open	High	Low	Close	Volume	Market Cap
0	2013-04-28	135.30	135.98	132.10	134.21	0	1500520000
1	2013-04-29	134.44	147.49	134.00	144.54	0	1491160000
2	2013-04-30	144.00	146.93	134.05	139.00	0	1597780000
3	2013-05-01	139.00	139.89	107.72	116.99	0	1542820000
4	2013-05-02	116.38	125.60	92.28	105.21	0	1292190000

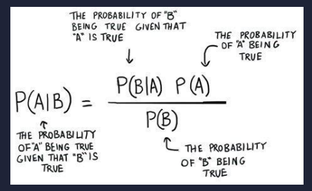
After

	Date	Close	Return_1d	Return_3d	SMA_7_diff	Volatility_7	Next_Return
0	2013-05-05	115.91	0.030311	0.101701	-0.024678	0.107695	-0.031145
1	2013-05-06	112.30	-0.031145	0.148849	-0.016957	0.099637	-0.007124
2	2013-05-07	111.50	-0.007124	-0.008889	0.010801	0.099961	0.018565
3	2013-05-08	113.57	0.018565	-0.020188	0.034147	0.081859	-0.007925
4	2013-05-09	112.67	-0.007925	0.003295	0.016091	0.069723	0.040206

Methodology: Classical Models Baseline

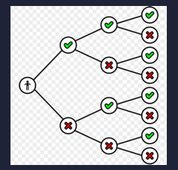
Our objective was to establish a baseline using established statistical and machine learning techniques.

Models tested



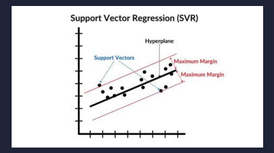
Gaussian Naive Bayes

To test if features are linearly separable.



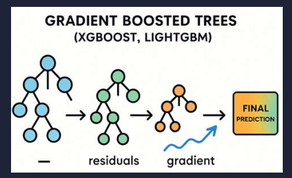
Decision Trees

Captures non-linear relationships and interactions.



Support Vector Regression

High-dimensional mapping for price regression.



Gradient Boosting

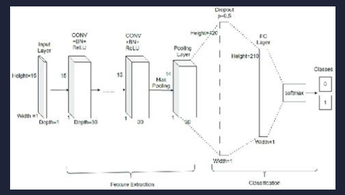
Ensemble method used for its high predictive power in tabular data.

Methodology: DL Models Experiments

Our objective was to capture local and long-term temporal dependencies in return sequences.

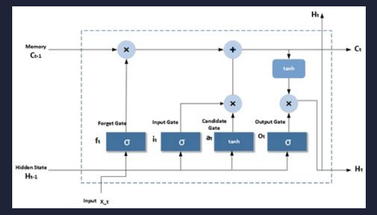
Optimisation: Implemented with Early Stopping (monitoring validation loss) and a 7-day lookback window in PyTorch.

Models tested



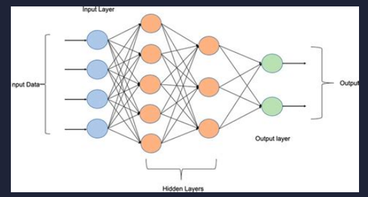
1D-CNN

Used for extracting local spatial-temporal features from price window



LSTM & GRU

Recurrent units designed to handle the "vanishing gradient" problem in long-term financial series



MLP Baseline

A flat dense network to measure the "sequencing gains."

Results: Metrics, SOTA and backtesting

Three complementary lenses

Success Metrics

Directional Accuracy (Signal Validity):

$$\mathrm{DA} = \frac{\text{Correct Sign Predictions}}{N} \times 100\%$$

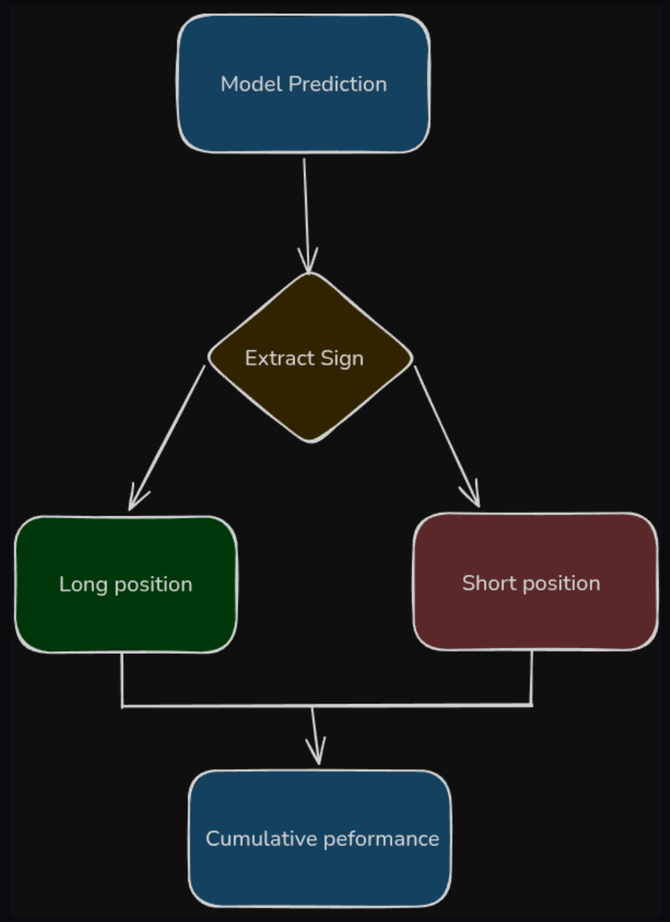
- **Goal:** Confirm if the predicted price movement direction is correct.
- **Benchmarking:** A ~55% accuracy is the industry standard for "State-of-the-Art" in daily markets. BitVision's Ridge model reaches ~63%, a substantial predictive edge over random noise.

Annualised Sharpe Ratio (Risk Efficiency):

$$S = \sqrt{365} \times \frac{\mu}{\sigma}$$

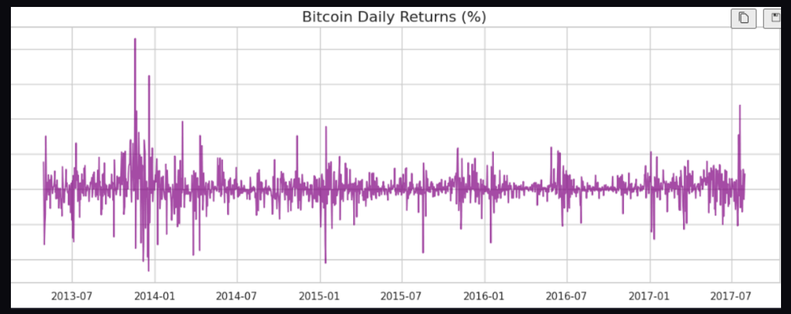
- **Goal:** Measure the consistency of returns relative to market volatility
- **Benchmarking:** A value > 1.0 is considered professional grade. Our final model achieved 2.94, signifying high stability with low comparative drawdown.

Backtesting

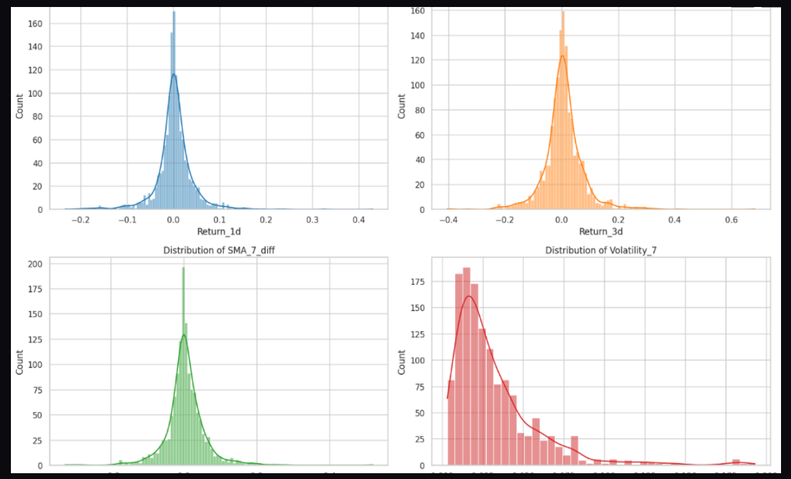


Results: EDA findings

The historical price chart illustrates Bitcoin's massive non-linear growth and inherent non-stationarity. Because raw prices exhibit a "random walk" with shifting means and variances, they are unsuitable for a direct linear prediction model. This visual confirms that we must focus on price dynamics rather than price levels to build a mathematically sound forecasting pipeline.



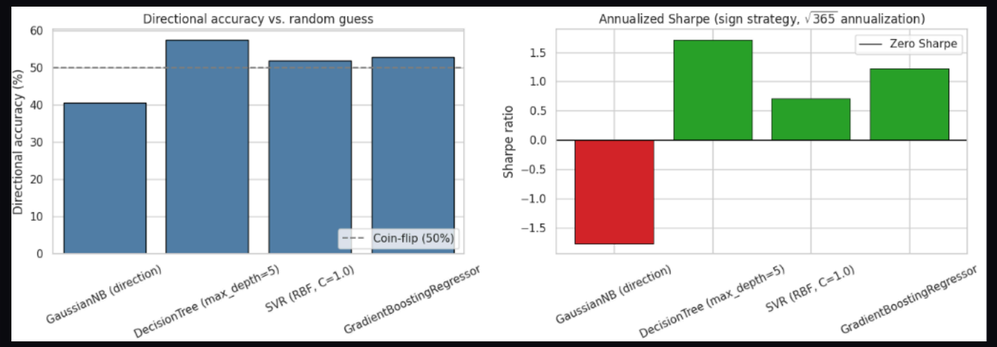
Converting prices to percentage returns reveals a stationary distribution centered near zero, which is the "signal" our model targets. Crucially, the chart shows distinct "clusters" where high-intensity spikes are followed by further volatility. This observation justifies our use of a 7-day rolling standard deviation, as it proves that past risk levels are a strong indicator of near-term market uncertainty.

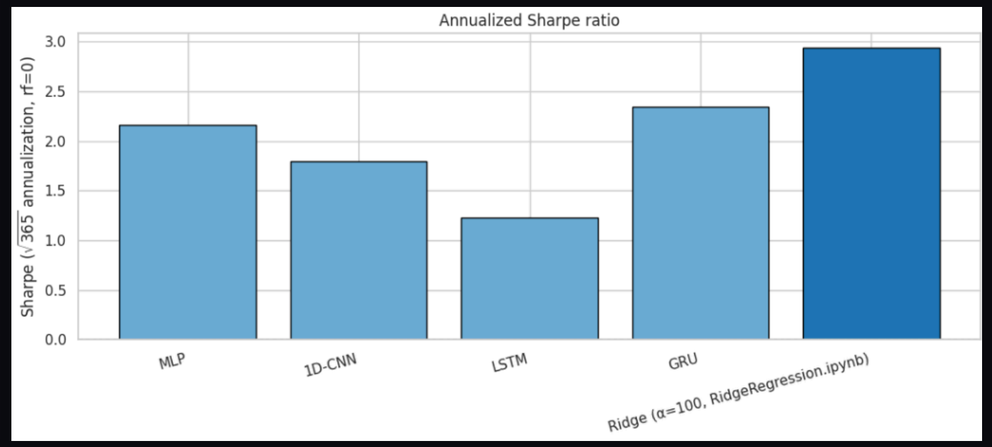


The distribution plots of our four features (Return_1d, Return_3d, SMA_7_diff, and Volatility_7) show deep "fat tails" (leptokurtosis), meaning extreme outlier events occur frequently. These outliers can easily skew a standard linear regression. This visualization is the primary evidence for choosing Ridge Regularization, which penalizes large coefficients to ensure the model remains robust and does not overfit to these rare but extreme market swings.

Results: Final Summary

Model	RMSE	Dir. acc. %	Ann. Sharpe
GaussianNB (direction)	70.3416	40.6452	-1.7707
DecisionTree (max_depth=5)	70.4188	57.4194	1.7195
SVR (RBF, C=1.0)	78.4756	51.9355	0.7140
GradientBoostingRegressor	68.8651	52.9032	1.2196
MLP	69.9042	58.7459	2.1656
1D-CNN	72.1195	59.0759	1.7929
LSTM	72.0871	58.7459	1.2282
GRU	72.7683	59.4059	2.3474
Ridge (α=100, RidgeRegression.ipynb)	68.9133	63.5484	2.9358

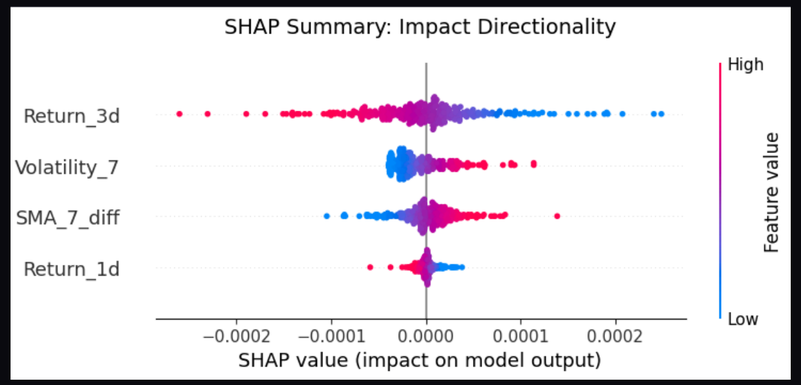




Results: Explainability and Analytics Dashboard

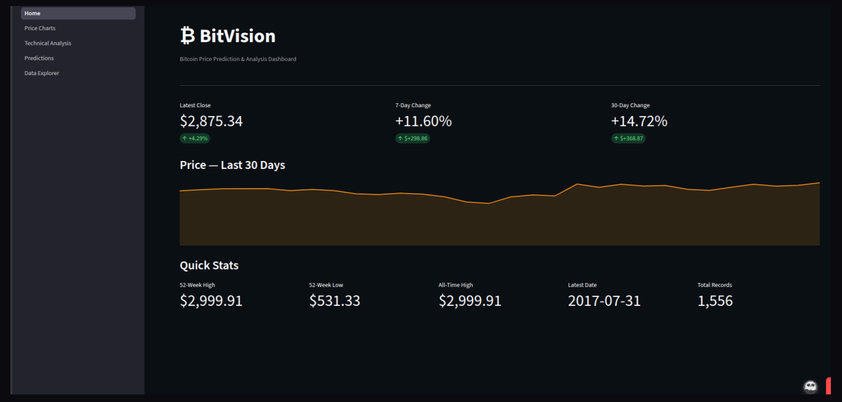
Explainability

- **SHAP Analysis:** We used SHAP (SHapley Additive exPlanations) to interpret the Ridge model. It reveals that the 3-day momentum is the strongest directional anchor.
- **Signal Magnitudes:** SHAP values confirm that the model relies on subtle signals (impacts of ~0.2% per day). While minor, these aggregate into a statistically dominant equity curve.
- **Robust Interpretability:** By using high Alpha (100.0), we ensure that the SHAP importance remains stable. Unlike Deep Learning "black boxes," we can precisely quantify why the model predicted a bull or bear move for the next day.



Streamlit Dashboard

Deployed!
Check it out: https://bitvision-iitj.streamlit.app/



Roles & Responsibilities

Akhil Dhyani
EDA
Data cleaning
Feature engineering
Model explainability

Divyansh Yadav
Final model pipeline (Ridge regression)
Streamlit application development
Cloud deployment

Harshit
Classical machine learning model experiments (GaussianNB, DecisionTree, SVR, GradientBoostingRegressor)

Gaurang Goyal
Deep learning model experiments (MLP, CNN, LSTM, GRU)

Thank You